

Ayoub GHRISS

Efficient ML, Sparsity & Quantization, GPU Systems

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EDUCATION

PhD, Machine Learning University of Colorado, Boulder Advisor: Claire Monteleoni. Expected graduation: Summer 2026. Thesis: Differentiable graph partitioning and sparse training for frugal machine learning.	2019 – 2026
Research Master – Mathematics, Computer Vision, and Machine Learning (MVA) École Normale Supérieure Paris-Saclay Reinforcement Learning, Probabilistic Graphical Models, Statistical Learning Computer Vision, Deep Learning, Speech Recognition, Audio Signal Processing	Oct 2017 – Sep 2018
Diplôme d'Ingénieur – Polytechnicien cycle École Polytechnique Polytechnician Engineer Cycle with Major in Applied and Computational Mathematics. Stochastic Calculus, Monte Carlo methods, Advanced Machine Learning, Random Models of survival.	Sep 2014 – Sep 2017
B.Sc. Economics and Applied Mathematics ENSAE ParisTech Data Analysis, Econometrics, Microeconomics, Macroeconomics	Sep 2013 – Jun 2014

LANGUAGES

Arabic	Native
French	Native
English	Fluent
Japanese	Intermediate
Chinese	Elementary

TECHNICAL SKILLS

ML / DL	Python	PyTorch
Scikit-learn	SciPy	
GPU & Systems	CUDA	Triton
C/C++	Linux	Git
Distributed	AWS	PySpark
Horovod	MPI	

WORK EXPERIENCE

Teaching Assistant CU Boulder, Computer Science Graduate Machine Learning (CSCI 5266 / 5622): Fall '21, Fall '23, Spring '26. Undergraduate Machine Learning (CSCI 4266), '22-'23; guest instructor for the Reinforcement Learning module. Tutorials on Small Language Models (Fall '23). Outstanding Teaching Assistant Award, '22.	Aug 2021 – Present	Applied Scientist Intern Amazon Alexa Paralinguistics, Cambridge Speaker identification and paralinguistic speech-emotion modeling on conversational audio.	May – Aug 2022
Graduate Research Assistant CIRES, CU Boulder GNN-based emulators for climate-model dynamics on refinable spherical meshes; released as the <i>Sphedron</i> library.	Jan 2024 – Dec 2025	Applied Scientist Intern Amazon Alexa Edge-ML, Cambridge Sentiment-aware ASR pre-training for on-device speech-emotion recognition (ICASSP '22).	May – Aug 2021
Visiting Researcher Inria, Paris Deep learning for optimization over discrete sets.	Jun – Aug 2023	Software Engineer, PhD Intern Google, Boulder Deep learning for air-quality prediction from satellite imagery (Google Earth Engine).	Jun – Aug 2020
		Research Intern RIKEN Advanced Intelligence Project, Tokyo Hierarchical deep reinforcement learning with options and learned state representations.	Apr – Sep 2018

PUBLICATIONS

Preprint	S ³ : Structured Sparsity Specification Ghriss, A.
AISTATS '26	Beyond Spectral Clustering: Probabilistic Cuts for Differentiable Graph Partitioning Ghriss, A. Outstanding Paper Award, CU Boulder CS Dept.
MusML ICML '25	Crowdsourced Information Authentication: A Graph-based Model from the Science of Hadith Ghriss, A. Oral presentation.
AISTATS '25	Deep Clustering via Probabilistic Ratio-Cut Optimization Ghriss, A., Monteleoni, C. Outstanding Paper Award, CU Boulder CS Dept.
ASCE '24	Predicting Serious Injury and Fatality Exposure using Machine Learning in Construction Projects Oguz Erkal, E.D., Hallowell, M.R., Ghriss, A., Bhandari, S.
ICASSP '22	Sentiment-aware ASR Pre-training for Enhanced Speech Emotion Recognition Ghriss, A., Yang, B., Rozgic, V., Shriberg, E., Wang, C.
NeuroImage '22	Insights from an Autism Imaging Biomarker Challenge: Promises and Threats to Biomarker Discovery Traut, N., et al. (incl. Ghriss, A.) Best ML model among participants.
RIKEN AIP '18	Reinforcement Learning with Options and State Representation Ghriss, A., Sugiyama, M., Lazaric, A.

SELECTED OPEN-SOURCE SOFTWARE

PGCuts github.com/ayghri/pgcuts • pgcuts.readthedocs.io
Differentiable graph cuts with GPU-accelerated hypergeometric kernels; sklearn-compatible API.

SparseKit github.com/ayghri/sparsekit • sparsekit.readthedocs.io
Structured sparsity specification (S^3) toolkit for carving LLM weights under user-defined patterns.

Qwantize github.com/ayghri/qwantize • qwantize.readthedocs.io
Optimal quantization for block-scaled low-precision formats (NVFP4, MXFP4).

Tritonix github.com/ayghri/tritonix
Triton GPU kernels with an optimized autotuning framework.

Sphedron github.com/ayghri/sphedron • sphedron.readthedocs.io
Refinable polyhedral meshes on the sphere for GNN-based geospatial ML.

SERVICE & TALKS

- **Reviewer:** ICLR 2025 • ICML 2025, 2026 • NeurIPS 2025
- **Tutorial Chair:** "Associative Memories", ICML 2025
- **Oral Presentation:** *Crowdsourced Information Authentication: A Graph-based Model from the Science of Hadith*, MusML Workshop, ICML 2025
- **Workshop Instructor:** *GPU Programming Workshop* — CS Graduate Professional Development Club, CU Boulder, Spring '26. Hands-on intro to parallel execution, thread/warp behavior, fused kernels, and Triton. github.com/ayghri/gpuwork
- **Guest Lectures:** *Introduction to Machine Learning* — ASEN 5307 Engineering Data Analysis Methods, CU Boulder (2 lectures in Fall '24, 2 in Fall '25)

AWARDS & HONORS

- **Outstanding Paper Award**, CU Boulder CS Dept. — for *Beyond Spectral Clustering: Probabilistic Cuts for Differentiable Graph Partitioning* (AISTATS '26).
- **Outstanding Paper Award**, CU Boulder CS Dept. — for *Deep Clustering via Probabilistic Ratio-Cut Optimization* (AISTATS '25).
- **Best Work-in-Progress Award**, CU Boulder CS Dept. Expo, Fall '25 — for work on structured sparse training (S^3).
- **Outstanding Teaching Assistant Award**, CU Boulder CS Dept., '22.